

Contents

1	Measure Theory	2
1.1	Outer Measure	2
1.2	Carathéodory Outer Measure	11
1.3	Lebesgue Measure	18
1.4	The Contor Set	32

Measure Theory

1.1 Outer Measure

Definition 1.1.1 ► Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an **outer measure** on X if the following conditions are satisfied:

- (i) $\varphi(\emptyset) = 0$,
- (ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,
- (iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,
- (iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X .

Example 1.1.2 ► Outer Measure

- (i) In an arbitrary set X , define $\varphi(A) = 1$ if A is nonempty and $\varphi(\emptyset) = 0$.
- (ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A .
- (iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$
- (iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A .
- (v) Select a fixed x_0 in an arbitrary set X , and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the **Dirac measure** concentrated at x_0 .

Definition 1.1.3

Let φ be an outer measure on a set X . A set $E \subset X$ is called **φ -measurable** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ -measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Note.

也就是说, E 是可测的, 当且仅当任意集合 A 都可以被 E 按外测度被良好地拆解为 $A \cap E$ 和 $A \setminus E$.

Lemma 1.1.4

A set $E \subset X$ is φ -measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Note.

也就是说 E 可测, 当且仅当落在 E 的内, 外的集合在外测度意义下是无关的.

Proof. For Necessity, if E is φ -measurable, then for all such P, Q ,

$$\varphi(P \cup Q) = \varphi((P \cup Q) \cap E) + \varphi((P \cup Q) - E) = \varphi(P) + \varphi(Q)$$

For sufficiency, let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi(A) = \varphi(P \cup Q) = \varphi(P) + \varphi(Q) = \varphi(A \cap E) + \varphi(A \setminus E)$$

□

Theorem 1.1.5

Suppose φ is an outer measure on an arbitrary set X . Then the following four statements hold:

- (i) E is φ -measurable whenever $\varphi(E) = 0$.
- (ii) $\emptyset$ and X are φ -measurable.
- (iii) $E_1 - E_2$ is φ -measurable whenever E_1 and E_2 are φ -measurable.
- (iv) If $\{E_i\}$ is a **countable collection of disjoint φ -measurable sets**, then

$\bigcup_{i=1}^{\infty} E_i$ is φ -measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c),$$

where

$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus

$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately from lemma 1.1.4.

3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.

$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$

Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implies that

$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$

Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have

$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$

Finally, we have

$$\begin{aligned} \varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2) \\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) \\ &= \varphi(P \cup Q) \end{aligned}$$

4. Let $S_k = \bigcup_{i=1}^k E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ -measurable and that

$$\varphi(A) \geq \sum_{i=1}^k \varphi(A \cap E_i) + \varphi(A \cap S_k^c), \quad \forall A \subseteq X$$

Then

$$\begin{aligned} \varphi(A) &= \varphi(A \cap E_{k+1}) + \varphi(A \cap E_{k+1}^c) \\ &= \varphi(A \cap E_{k+1}) + \varphi(A \cap E_{k+1}^c \cap S_k^c) + \varphi(A \cap E_{k+1}^c \cap S_k) \\ &= \varphi(A \cap E_{k+1}) + \varphi(A \cap S_{k+1}^c) + \varphi(A \cap S_k) \end{aligned}$$

Replace A by $A \cap S_k$, we have

$$\varphi(A \cap S_k) \geq \sum_{i=1}^k \varphi(A \cap S_k \cap E_i) + \varphi(A \cap S_k \cap S_k^c) = \sum_{i=1}^k \varphi(A \cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1} \varphi(A \cap E_{k+1}) + \varphi(A \cap S_{k+1}^c)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A \cap S_{k+1}) + \varphi(A \cap S_{k+1}^c)$$

this, in turn, implies that S_{k+1} is φ -measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^k \varphi(A \cap E_i) + \varphi(A \cap S_k^c)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \geq \varphi(A \cap S) + \varphi(A \cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable. The subadditivity gives that

$$\varphi(A) \leq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Finally, we have

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Replace A by S , we can get

$$\varphi(S) = \sum_{i=1}^{\infty} \varphi(E_i)$$

□

Lemma 1.1.6

Let $\{E_i\}$ be a sequence of arbitrary sets. Then there exists a sequence of disjoint sets $\{A_i\}$ such that each $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ -measurable, and so is A_i .

Proof. Let $S_k = \bigcup_{i=1}^k E_i$. Let $A_k = S_k \setminus S_{k-1}$, $k \geq 1$, where $S_0 = \emptyset$. Then

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} (S_k \setminus S_{k-1}) = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ -measurable. $E_{i+1} \cup E_i = (E_{i+1} - E_i) \cup E_i$ is measurable follows from the (ii), (iii) of the above theorem. Inductively, every $S_k = (S_{k-1} \setminus E_k) \cup E_k$ is measurable. Then each A_k is measurable. □

Theorem 1.1.7

If $\{E_i\}$ is a sequence of φ -*measurable* sets in X , then $\cup_{i=1}^{\infty} E_i$ and $\cap_{i=1}^{\infty} E_i$ are φ -*measurable*.

Proof. By the lemma above, there exists a sequence of disjoint φ -measurable sets $\{A_i\}$ such that each $A_i \subseteq E_i$, and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i$$

It follows from Theorem 1.1.5 that $\bigcup_{i=1}^{\infty} A_i$ is measurable, then $\bigcup_{i=1}^{\infty} E_i$ is as well. Since X is measurable, $E_i^c = X \setminus E_i$ is measurable, then $\bigcup_{i=1}^{\infty} E_i^c$ is measurable. Thus

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \bigcup_{i=1}^{\infty} E_i^c$$

is measurable. □

Definition 1.1.8

A nonempty collection Σ of sets E satisfying the following two conditions is called a σ -*algebra*:

- (i) if $E \in \Sigma$, then $E^c \in \Sigma$;
- (ii) $\cup_{i=1}^{\infty} E_i \in \Sigma$ if each E_i is in Σ .

Definition 1.1.9

In a topological space, the elements of the smallest σ -algebra that contains all open sets are called **Borel sets**. The term “smallest” is taken in the sense of inclusion.

Proof. We need to show that the “smallest” exists. □

Corollary 1.1.10

If φ is an *outer measure* on an arbitrary set X , then the class of φ -measurable sets forms a σ -*algebra*.

Proof. Follows immediately from Theorem 1.1.5 and Theorem 1.1.7. □

Corollary 1.1.11

Suppose φ is an outer measure on X and $\{E_i\}$ a countable collection of φ -measurable sets.

(i) If $E_1 \subset E_2$ with $\varphi(E_1) < \infty$, then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1).$$

(ii) (Countable additivity) If $\{E_i\}$ is a disjoint sequence of sets, then

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

(iii) (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i , then

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \varphi\left(\lim_{i \rightarrow \infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i).$$

(iv) (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i , and if $\varphi(E_{i_0}) < \infty$ for some i_0 , then

$$\varphi\left(\bigcap_{i=1}^{\infty} E_i\right) = \varphi\left(\lim_{i \rightarrow \infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i).$$

(v) If $\{E_i\}$ is any sequence of φ -measurable sets, then

$$\varphi\left(\liminf_{i \rightarrow \infty} E_i\right) \leq \liminf_{i \rightarrow \infty} \varphi(E_i).$$

(vi) If

$$\varphi\left(\bigcup_{i=i_0}^{\infty} E_i\right) < \infty$$

for some positive integer i_0 , then

$$\varphi\left(\limsup_{i \rightarrow \infty} E_i\right) \geq \limsup_{i \rightarrow \infty} \varphi(E_i).$$

Proof. (v) Let $A_j = \bigcap_{k=j}^{\infty} E_k$, then

$$\liminf_{i \rightarrow \infty} E_i = \lim_{j \rightarrow \infty} A_j = \bigcup_{j=1}^{\infty} A_j$$

It follows from (iv) that

$$\varphi\left(\liminf_{i \rightarrow \infty} E_i\right) = \lim_{j \rightarrow \infty} \varphi(A_j)$$

Since $A_j \subseteq E_j$, we have $\varphi(A_j) \leq \varphi(E_j)$, then

$$\lim_{j \rightarrow \infty} \varphi(A_j) \leq \liminf_{i \rightarrow \infty} \varphi(E_i)$$

Thus

$$\varphi\left(\liminf_{i \rightarrow \infty} E_i\right) \leq \liminf_{i \rightarrow \infty} \varphi(E_i)$$

□

Definition 1.1.12

- An outer measure φ on a topological space X is called a **Borel outer measure** if all Borel sets are φ -measurable.
- A Borel outer measure is **finite** if $\varphi(X)$ is finite.
- An outer measure φ on a set X is called **regular** if for each $A \subseteq X$ there exists a φ -measurable set $B \supseteq A$ such that $\varphi(B) = \varphi(A)$.
- A **Borel regular outer measure** is a Borel outer measure such that for each $A \subseteq X$, there exists a Borel set B such that $\varphi(B) = \varphi(A)$.
- A **Radon outer measure** is a Borel regular outer measure that is finite on compact sets.

Exercises for Outer Measure

Exercise 1.1.13

If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A .

Let $\varphi_\varepsilon(A) := \varphi(A)$ denote the dependence on ε , and define

$$\psi(A) := \lim_{\varepsilon \rightarrow 0} \varphi_\varepsilon(A)$$

What is $\psi(A)$ and what are the corresponding ψ -measurable sets.

Proof. If there are only finite points in A . Let

$$l := \min \{d(x, y) : x, y \in A, x \neq y\}$$

Then for any $x, y \in A, x \neq y$. If B_1, B_2 are ball of radius $\frac{l}{2}$ such that $x \in B_1, y \in B_2$. Then $B_1 \cap B_2 = \emptyset$. But each point in A must covered by one of such ball, which shows that $\psi(A) \geq |A|$. Since it is obvious that $\psi(A) \leq |A|$, then $\psi(A) = |A|$.

Now suppose that A is a infinite set. If $\psi(A) < \infty$, say $\psi(A) = n$. Then for any sufficiently small ε , there are balls $B_1, \dots, B_n$ with radius ε cover A . Take any $n + 1$ different points $p_1, \dots, p_{n+1}$ of A . Let $\varepsilon = \frac{1}{2} \min \{d(p_i, p_j) : i \neq j\}$, then $B_1, \dots, B_n$ cannot cover these $n + 1$ points, thus cannot cover A , which is a contradiction. Thus we have $\psi(A) = \infty$.

We find that ψ is just the counting measure on X . Take any $E \subseteq X$, and take any $A \subseteq X$, if A is finite, then $A \cap E, A - E$ are as well, and $|A| = |A \cap E| + |A - E|$, that is $\psi(A) = \psi(A \cap E) + \psi(A - E)$. If A is infinite, then one of the $A \cap E$ and $A - E$ is as well. Then

$$\psi(A) = \infty = \psi(A \cap E) + \psi(A - E)$$

The above shows that E is ψ -measurable. Thus all subset of X are ψ -measurable. $\square$

Exercise 1.1.14

It was shown that $\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1)$ if $E_1 \subseteq E_2$ are φ -measurable with $\varphi(E_1) < \infty$. Prove that this result remains true if E_2 is not assumed to be φ -measurable.

Proof. Since E_1 is measurable,

$$\varphi(E_2) = \varphi(E_2 \cap E_1) + \varphi(E_2 - E_1)$$

but

$$\varphi(E_2 \cap E_1) = \varphi(E_1)$$

which completes the proof. □

1.2 Carathéodory Outer Measure

Definition 1.2.1 ▶ Carathéodory outer measure

An outer measure φ defined on a metric space (X, ρ) is called a **Carathéodory outer measure** if

$$\varphi(A \cup B) = \varphi(A) + \varphi(B)$$

whenever A, B are arbitrary subsets of X with $d(A, B) > 0$. Where

$$d(A, B) := \inf\{\rho(a, b) : a \in A, b \in B\}$$

Theorem 1.2.2

If φ is a Carathéodory outer measure on a metric space X , then all closed sets are φ -measurable.

Proof sketch.

用一系列集合近似代替 C 是标准的想法. 证明的核心要素有: 1. C 是闭集, 使得按距离向外逼近的行为是良好的. 2. 距离这个属性与 Carathéodory 条件是相适应的. 此外, 将命题转化为极限

$$\lim_{i \rightarrow \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

是标准的流程. 要证明该极限成立, 则需要一定的技巧. 这里将 $A \setminus C_i$ 和 $A \setminus C$ 相差的中间部分进行细分量化, 将原极限转化为

$$\lim_{i \rightarrow \infty} \sum_{j=i}^{\infty} \varphi(T_j)$$

的极限问题, 从而转化为级数的收敛性问题. 将集合隔一个分组, 变成两组两两相隔正距离的集合, 以便套用 Carathéodory 条件.

Proof. Let $C \subseteq X$ be any closed sets. Take $A \subseteq X$, only need to show that

$$\varphi(A) \geq \varphi(A \cap C) + \varphi(A \setminus C)$$

We assume that $\varphi(A) < \infty$. Define

$$C_i := \left\{ x : d(x, C) \leq \frac{1}{i} \right\}$$

Since $d(A \setminus C_i, A \cap C) > 0$,

$$\varphi(A) \geq \varphi((A \setminus C_i) \cup (A \cap C)) = \varphi(A \setminus C_i) + \varphi(A \cap C)$$

Only need to show that

$$\lim_{i \rightarrow \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

Define

$$T_i := \left\{ x \in A : \frac{1}{i+1} < d(x, C) \leq \frac{1}{i} \right\}$$

, then

$$A \setminus C \subseteq (A \setminus C_i) \cup \bigcup_{j=i}^{\infty} T_j$$

Thus

$$\varphi(A \setminus C) \leq \varphi(A \setminus C_i) + \sum_{j=i}^{\infty} \varphi(T_j)$$

Once we have $\sum_{j=1}^{\infty} \varphi(T_j) < \infty$, then $\lim_{i \rightarrow \infty} \sum_{j=i}^{\infty} \varphi(T_j) = 0$, the conclusion follows from it. To show this, note that $d(T_i, T_j) > 0$ whenever $|i - j| \geq 2$. Then it by induction follows from Carathéodory condition that

$$\sum_{j=1}^{\infty} \varphi(T_{2j}) \leq \varphi(A) < \infty$$

$$\sum_{j=1}^{\infty} \varphi(T_{2j-1}) \leq \varphi(A) < \infty$$

Thus $\sum_{j=1}^{\infty} \varphi(T_j) < \infty$, which completes the proof. □

Theorem 1.2.3

Suppose $\mathcal{F}$ is a family of subsets of a topological space X that contains all open and all closed subsets of X . Suppose also that $\mathcal{F}$ is closed under countable unions and countable intersections. Then $\mathcal{F}$ contains all Borel sets; that is, $\mathcal{B} \subset \mathcal{F}$.

Note.

可以将包含 $\mathcal{B}$ 描述成对以下三种集合的封闭性: 1. 开集 2. 补集 3. 可数并. 如果我们事先没有对补集的封闭性, 可以通过加强 1.3. 的条件, 使得当我们强制使得补集封闭后, 得到的新集族依旧满足 1.3. 具体地, 我们希望原集族对于开集的补集, 也就是闭集也封闭; 对于可数并的补集, 也就是可数交也封闭.

Proof. Let

$$\mathcal{H} = \mathcal{F} \cap \{A : A^c \in \mathcal{F}\}$$

Observe that $\mathcal{H}$ contains all closed sets. Moreover, it is easily seen that $\mathcal{H}$ is closed under complementation and countable unions. Thus $\mathcal{H}$ is a σ -algebra that contains the open sets and therefore contains all Borel sets. $\square$

Theorem 1.2.4

If φ is a Carathéodory outer measure on a metric space X , then the Borel sets of X are φ -measurable.

Proof. Since the collection of φ -measurable sets is itself a σ -algebra, it is closed under complements and countable union and countable intersection. Theorem 1.2.2 gives that φ -measurable sets contains closed sets, thus contains open sets as well. Then the conclusion follows from Theorem 1.2.3. $\square$

Exercises for this section

Exercise 1.2.5 ▶ Smallest σ -algebra containing open sets

Prove that in every topological space X , there exists a smallest σ -algebra that contains all open sets in X . That is, prove that there is a σ -algebra Σ that contains all open sets and has the property that if Σ_1 is another σ -algebra containing all open sets, then $\Sigma \subset \Sigma_1$. In particular, for $X = \mathbb{R}^n$, note that there is a smallest σ -algebra that contains all the closed sets in $\mathbb{R}^n$.

Proof. Let

$$\Sigma := \bigcap_{\mathcal{M}_\alpha \text{ is a } \sigma \text{ algebra containing } X} \mathcal{M}_\alpha$$

Then it is obvious that Σ is closed under complement, countable union, and contains all open sets. Thus Σ is a σ -algebra containing all open sets. $\square$

Exercise 1.2.6

In a topological space X the family of Borel sets, $\mathcal{B}$, is by definition the σ -algebra generated by the closed sets. The method below is another way of describing the Borel sets using transfinite induction. You are to fill in the necessary steps:

1. For an arbitrary family $\mathcal{F}$ of sets, let

$$\mathcal{F}^* = \left\{ \bigcup_{k=1}^{\infty} E_k : \text{where either } E_i \in \mathcal{F} \text{ or } \tilde{E}_i \in \mathcal{F} \text{ for all } i \in \mathbb{N} \right\}.$$

Let Ω denote the smallest uncountable ordinal. We will use transfinite induction to define a family $\mathcal{E}_\alpha$ for each $\alpha < \Omega$.

2. Let $\mathcal{E}_0 :=$ all closed sets $:= \mathcal{K}$. Now choose $\alpha < \Omega$ and assume that $\mathcal{E}_\beta$ has been defined for each β such that $0 \leq \beta < \alpha$. Define

$$\mathcal{E}_\alpha := \left(\bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta \right)^*$$

and define

$$\mathcal{A} := \bigcup_{0 \leq \alpha < \Omega} \mathcal{E}_\alpha.$$

3. Show that each $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. Show that $\mathcal{A} \subset \mathcal{B}$.
5. Now show that $\mathcal{A}$ is a σ -algebra to conclude that $\mathcal{A} = \mathcal{B}$.
 - (a) Show that $\emptyset, X \in \mathcal{A}$.
 - (b) Let $A \in \mathcal{A} \implies A \in \mathcal{E}_\alpha$ for some $\alpha < \Omega$. Show that this implies $\tilde{A} \in \mathcal{E}_\alpha^* \subset \mathcal{E}_\beta$ for every $\beta > \alpha$.
 - (c) Conclude that $\tilde{A} \in \mathcal{A}$ and thus conclude that $\mathcal{A}$ is closed under complementation.
 - (d) Now let $\{A_k\}$ be a sequence in $\mathcal{A}$. Show that $\bigcup_{k=1}^{\infty} A_k \in \mathcal{A}$. (Hint: Each A_k is in $\mathcal{A}_{\alpha_k}$ for some $\alpha_k < \Omega$. We know that there is $\beta < \Omega$ such that $\beta > \alpha_k$ for each $k \in \mathbb{N}$.)

Proof. 3. Since $\mathcal{E}_0 \subseteq \mathcal{B}$, only need to show that if $\mathcal{E}_\beta \subseteq \mathcal{B}$ for all $0 \leq \beta < \alpha$, then $\mathcal{E}_\alpha \subseteq \mathcal{B}$. Take any $E \in \mathcal{E}_\alpha$, then there are $E_1, \dots, E_k, \dots$, such that $E_i \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, or $E_i^c \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, and $E = \bigcup_{k=1}^{\infty} E_k$. We suppose that $E_i \in \mathcal{E}_{\beta_i}$ or $E_i^c \in \mathcal{E}_{\beta_i}$. Since $\mathcal{E}_{\beta_i} \subseteq \mathcal{B}$ for all β_i , and since $\mathcal{B}$ is closed under complement, then $E_i \in \mathcal{B}$. Since $\mathcal{B}$ is closed under countable union, $E = \bigcup_{k=1}^{\infty} E_k \in \mathcal{B}$. Thus $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. We have shown that $\mathcal{E}_\alpha \subseteq \mathcal{B}$ for all $\alpha < \Omega$, then it is immediate that $\mathcal{A} \subseteq \mathcal{B}$.

5. (a) Since $\emptyset \in \mathcal{E}_0$ and $X = (\emptyset)^c \in \mathcal{E}_1 = (\mathcal{E}_0)^*$, then $\emptyset, X \in \mathcal{A}$.

(b) Write

$$\mathcal{E}_\alpha^* = \left\{ \bigcup_{k=1}^{\infty} E_k : \text{where either } E_i \in \mathcal{E}_\alpha \text{ or } E_i^c \in \mathcal{E}_\alpha \text{ for all } i \in \mathbb{N} \right\}$$

We take $E_1 = A^c$, then $E_1^c = A \in \mathcal{E}_\alpha$, and take $E_2 = \dots = \emptyset$. Then $A^c = \bigcup_{k=1}^{\infty} E_k \in \mathcal{E}_\alpha^*$. If $\beta > \alpha$, then

$$\mathcal{E}_\alpha \subseteq \bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma$$

Let $*$ act on both side, then it follows from the definition of $*$ that

$$(\mathcal{E}_\alpha)^* \subseteq \left(\bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma \right)^* = \mathcal{E}_\beta$$

Thus $A^c \in \mathcal{E}_\alpha^* \subseteq \mathcal{E}_\beta$ for every $\beta > \alpha$.

(c) It is immediate from (b) that $A^c \in \mathcal{E}_\beta \subseteq \mathcal{A}$.

(d) Suppose that $A_k \in \mathcal{E}_{\alpha_k}$ for some $\alpha_k < \Omega$. Let

$$\gamma := \bigcup_{k \in \mathbb{N}} \alpha_k, \quad \beta := \gamma + 1$$

Then $\gamma, \beta < \Omega$, and $\alpha_1, \dots, \alpha_k < \beta$. Then

$$\bigcup_{k=1}^{\infty} A_k \in \left(\bigcup_{k=1}^{\infty} \mathcal{E}_{\alpha_k} \right)^* \subseteq \mathcal{E}_\beta \subseteq \mathcal{A}$$

□

Exercise 1.2.7 ► *

An outer measure φ on a space X is called σ -finite if there exists a countable number of sets A_i with $\varphi(A_i) < \infty$ such that $X \subseteq \bigcup_{i=1}^{\infty} A_i$. Assuming that φ is a σ -finite Borel regular outer measure on a metric space X , prove that $E \subseteq X$ is φ -measurable if and only if there exists an F_σ set $F \subseteq E$ such that $\varphi(E \setminus F) = 0$.

Proof. Let $E_i := E \cap A_i$. Then

$$E = \bigcup_{i=1}^{\infty} E_i, \quad \varphi(E_i) \leq \varphi(A_i) < \infty$$

Once we show that there is a closed set $F_i \subseteq E_i$, such that $\varphi(E_i \setminus F_i) = 0$ for all i , then let $F = \bigcup_{i=1}^{\infty} F_i \in F_\sigma$, it follows that

$$\varphi(E \setminus F) \leq \varphi\left(\bigcup (E \setminus F_i)\right) \leq \varphi\left(\bigcup (E_i \setminus F_i)\right) \leq \sum_i \varphi(E_i \setminus F_i) = 0$$

□

Exercise 1.2.8

Let φ be an outer measure on a space X . Suppose $A \subset X$ is an arbitrary set with $\varphi(A) < \infty$ such that there exists a φ -measurable set $E \supset A$ with $\varphi(E) = \varphi(A)$. Prove that $\varphi(A \cap B) = \varphi(E \cap B)$ for every φ -measurable set

B.

Proof.

$$\begin{aligned}
 \varphi(A) &= \varphi(E) = \varphi(E \cap B) + \varphi(E \setminus B) \\
 \varphi(A \cap B) &\geq \varphi(A) - \varphi(A \setminus B) = \varphi(E) - \varphi(A \setminus B) \\
 &= \varphi(E \cap B) + \varphi(E \setminus B) - \varphi(A \setminus B) \\
 &\geq \varphi(E \cap B)
 \end{aligned}$$

Since $A \cap B \subseteq E \cap B$, $\varphi(A \cap B) \leq \varphi(E \cap B)$. Finally, we have $\varphi(A \cap B) = \varphi(E \cap B)$ $\square$

Exercise 1.2.9

In $\mathbb{R}^2$, find two disjoint closed sets A and B such that $\mathbf{d}(A, B) = 0$. Show that this is not possible if one of the sets is compact.

Proof. 1. Let A be the graph of $y = \frac{1}{x}$. Let $B = \{(x, y) : y = 0\}$. Then $\left\{\left(n, \frac{1}{n}\right) : n \in \mathbb{N}\right\} \subseteq A$, $\{(n, 0) : n \in \mathbb{N}\} \subseteq B$, then

$$\mathbf{d}(A, B) \leq \inf \left\{ d\left(\left(n, \frac{1}{n}\right), (n, 0)\right) : n \in \mathbb{N} \right\} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

2. If A is compact, B is closed. Since $\mathbf{d}(A, B) = 0$, there are convergent sequences $\{a_n\} \subseteq A$ and $\{b_n\} \subseteq B$ such that $\lim_{n \rightarrow \infty} d(a_n, b_n) = 0$. Since A is bounded and closed, there is a cluster point $a \in A$ of $\{a_n\}$. Suppose that $\lim_{k \rightarrow \infty} a_{n_k} = a$, then $\lim_{k \rightarrow \infty} b_{n_k} = a \in B$. $A \cap B \neq \emptyset$, which is a contradiction. $\square$

Exercise 1.2.10

Let φ be an outer Carathéodory measure on $\mathbb{R}$ and let $f(x) := \varphi(I_x)$, where I_x is an open interval of fixed length centered at x . Prove that f is lower semicontinuous. What can you say about f if I_x is taken as a closed interval? Prove the analogous result in $\mathbb{R}^n$; that is, let $f(x) := \varphi(B(x, a))$, where $B(x, a)$ is the open ball with fixed radius a centered at x .

Exercise 1.2.11

In a metric space X , prove that $\text{dist}(A, B) = d(\bar{A}, \bar{B})$ for arbitrary sets $A, B \in X$.

Exercise 1.2.12

Let A be a non-Borel subset of $\mathbb{R}^n$ and define for each subset E ,

$$\varphi(E) = \begin{cases} 0 & \text{if } E \subset A, \\ \infty & \text{if } E \setminus A \neq \emptyset. \end{cases}$$

Prove that φ is an outer measure that is not Borel regular.

Exercise 1.2.13

Let $\mathcal{M}$ denote the class of φ -measurable sets of an outer measure φ defined on a set X . If $\varphi(X) < \infty$, prove that the family

$$\mathcal{F} := \{A \in \mathcal{M} : \varphi(A) > 0\}$$

is at most countable.

1.3 Lebesgue Measure

Theorem 1.3.1

Suppose each edge $I_k = [a_k, b_k]$ of an n -dimensional interval I is partitioned into α_k subintervals. The products of these intervals produce a partition of I into $\beta := \alpha_1 \cdot \alpha_2 \cdots \alpha_n$ subintervals I_i and

$$v(I) = \sum_{i=1}^{\beta} v(I_i).$$

Theorem 1.3.2

For each interval I and each $\epsilon > 0$, there exists an interval J whose interior contains I and

$$v(J) < v(I) + \epsilon.$$

Definition 1.3.3 ▶ Lebesgue outer measure

The **Lebesgue outer measure** of an arbitrary set $E \subseteq \mathbb{R}^n$, denoted by $\lambda^*(E)$, is defined by

$$\lambda^*(E) := \inf \left\{ \sum_{k=1}^{\infty} v(I_k) \right\},$$

where the infimum is taken over all countable collections of closed intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k.$$

Remark.

- It may be necessary at times to emphasize the dimension of the Euclidean space in which Lebesgue outer measure is defined. When clarification is needed, we will write $\lambda_n^*(E)$ in place of $\lambda^*(E)$.
- We have not proved that **Lebesgue outer measure** is actually a outer measure till now.

Theorem 1.3.4

For a closed interval $I \subseteq \mathbb{R}^n$, $\lambda^*(I) = v(I)$.

Proof. The inequality $\lambda^*(I) \leq v(I)$ holds, since I itself gives a element of $\left\{ \sum_{k=1}^{\infty} v(I_k) \right\}$ in the definition of $\lambda^*(I)$.

To prove the opposite inequality, choose $\varepsilon > 0$, and a sequence of closed intervals $I_1, I_2, \dots$ such that

$$I \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \leq \lambda^*(I) + \varepsilon$$

We can choose closed intervals J_k such that

$$v(J_k) \leq v(I_k) + \frac{\varepsilon}{2^k}, \quad I_k \subseteq J_k^\circ$$

Let $\mathcal{F} := \{J_k^\circ : k \in \mathbb{N}\}$. Then $\mathcal{F}$ is an open cover of I , since I is compact, it has a lebesgue number η . There is a finitely many partition $K_1, \dots, K_m$ of I such

that each with diameter less than η . Then K_i is contained in some element of $\mathcal{F}$, we say J_{k_i} although two K_i may be contained in a same J_{k_i} . Let N_m be the smallest number of J_{k_i} containing K_i , then

$$v(I) = \sum_{i=1}^m v(K_i) \leq \sum_{i=1}^{N_m} v(J_{k_i}) \leq \sum_{k=1}^{\infty} v(J_k) \leq \lambda^*(E) + 2\varepsilon$$

Since ε is arbitrary, $v(I) \leq \lambda^*(E)$. □

Theorem 1.3.5

Lebesgue outermeasure, λ^* , defined on $\mathbb{R}^n$ is a Carathéorody outer measure.

Proof. We first verify that λ^* is an outer measure. We only show the subadditivity of λ^* , the other conditions are obvious. Let $\{A_i\}$ be a countable collection of arbitrary subsets of $\mathbb{R}^n$. Denote $A := \bigcup_{i=1}^{\infty} A_i$. There is a countable family of closed intervals $\{I_j^{(i)}\}_{j \in \mathbb{N}}$ such that

$$A_i \subseteq \bigcup_{j=1}^{\infty} I_j^{(i)}, \quad \sum_{j=1}^{\infty} v(I_j^{(i)}) < \lambda^*(A_i) + \frac{\varepsilon}{2^i}$$

Collection of closed interval $\{I_j^{(i)}\}_{i,j \in \mathbb{N}}$ can be arranged to a sequence of closed intervals, such that

$$A \subseteq \bigcup_{i,j \in \mathbb{N}} I_j^{(i)}$$

Thus

$$\lambda^*(A) \leq \sum_{i,j} v(I_j^{(i)}) \leq \sum_{i=1}^{\infty} \lambda^*(A_i) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, the countable subadditivity of λ^* is established.

To show that λ^* is Carathéorody outer measure. We take any $\varepsilon > 0$, and take a countable collection of closed intervals $I_1, I_2, \dots$ such that

$$A \cup B \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \subseteq \lambda^*(A \cup B).$$

We can subdividing each I_k into smaller intervals if necessary, we may assume that each interval I_k which diameter less than $\mathbf{d}(A, B)$. Then $\{I_k\}$ can be separated into two subfamilies $\{I'_k\}$ and $\{I''_k\}$. The members in the former have empty intersection with B , and that in the later have empty intersection with A . $\{I_k\} = \{I'_k\} \sqcup \{I''_k\}$. Thus

$$A \subseteq \bigcup_{k=1}^{\infty} \{I'_k\}, \quad B \subseteq \bigcup_{k=1}^{\infty} \{I''_k\}$$

We have

$$\begin{aligned} \lambda^*(A) + \lambda^*(B) &\leq \sum_{k=1}^{\infty} \nu(I'_k) + \sum_{k=1}^{\infty} \nu(I''_k) \\ &= \sum_{k=1}^{\infty} \nu(I_k) \\ &< \lambda^*(A \cup B) + \varepsilon \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda^*(A) + \lambda^*(B) \leq \lambda^*(A \cup B)$$

□

Corollary 1.3.6

All Borel sets in $\mathbb{R}^n$ are Lebesgue measurable. In particular, each open set and each closed set are Lebesgue measurable.

Definition 1.3.7 ▶ Lebesgue Measure

Denote by λ the set function obtained by restricting λ^* to the family of Lebesgue measurable sets. λ is called **Lebesgue measure**.

Corollary 1.3.8

Let J be an open interval in $\mathbb{R}^n$. Then

$$\lambda(J) = \text{vol}(J)$$

Proof. There is a sequence of closed intervals I_k such that $J = \bigcup_{k=1}^{\infty} I_k$, then

$$\lambda(J) = \lim_{k \rightarrow \infty} \lambda(I_k) = \lim_{k \rightarrow \infty} \text{vol}(I_k) = \text{vol}(J)$$

□

Theorem 1.3.9

The following five conditions are equivalent for Lebesgue outer measure, λ^* , on $\mathbb{R}^n$:

1. $E \subseteq \mathbb{R}^n$ is λ^* -measurable.
2. For each $\varepsilon > 0$, there is an open set $U \supseteq E$ such that $\lambda^*(U \setminus E) < \varepsilon$.
3. There is a G_δ set $U \supseteq E$ such that $\lambda^*(U \setminus E) = 0$.
4. For each $\varepsilon > 0$, there is a closed set $F \subseteq E$ such that $\lambda^*(E \setminus F) < \varepsilon$.
5. There is an F_σ set $F \subseteq E$ such that $\lambda^*(E \setminus F) = 0$.

Proof. 1. $\implies$ 2. : We first assume that $\lambda^*(E) < \infty$. Take closed intervals $I_1, I_2, \dots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \lambda^*(E) + \frac{\varepsilon}{2}$$

Let I'_k be closed intervals whose interior contains I_k , and

$$v(I'_k) < v(I_k) + \frac{\varepsilon}{2^{k+1}}$$

Let

$$U := \bigcup_{k=1}^{\infty} (I'_k)^\circ$$

is an open set. Since U, E is λ -measurable,

$$\begin{aligned} \lambda^*(U \setminus E) &= \lambda(U \setminus E) = \lambda(U) - \lambda(E) \\ &= \lambda(U) - \lambda^*(E) \\ &< \sum_{k=1}^{\infty} v(I'_k) - \left(\sum_{k=1}^{\infty} v(I_k) - \frac{\varepsilon}{2} \right) \\ &\leq \varepsilon \end{aligned}$$

Now, note that $\mathbb{R}^n$ is σ -finite, we suppose that $A_1, A_2, \dots$ are measurable sets

such that

$$\mathbb{R}^n = \bigcup_{j=1}^{\infty} A_j, \quad \lambda(A_j) < \infty, \forall j \in \mathbb{Z}_{>0}$$

Then $E_j := E \cap A_j$ is a λ^* -measurable set with $\lambda(E_j) < \infty$ for each j . For each $\varepsilon > 0$, and each $j \in \mathbb{Z}_{>0}$, there is an open set $U_j \supseteq E$ such that

$$\lambda^*(U_j \setminus E_j) < \frac{\varepsilon}{2^j}$$

Let $U := \bigcup_{j=1}^{\infty} U_j$, then

$$\lambda^*(U \setminus E) \leq \sum_{j=1}^{\infty} \lambda^*(U_j \setminus E) \leq \sum_{j=1}^{\infty} \lambda^*(U_j \setminus E_j) \leq \varepsilon$$

which completes the proof.

2. $\implies$ 3. : For each $n \in \mathbb{Z}_{>0}$, there is a open set U_n , such that

$$\lambda^*(U_n \setminus E) < \frac{1}{n}$$

Let

$$U := \bigcap_{n=1}^{\infty} U_n$$

then U is a G_δ set, such that

$$\lambda^*(U \setminus E) \leq \lambda^*(U_n \setminus E) < \frac{1}{n}, \quad \forall n.$$

That is $\lambda^*(U \setminus E) = 0$.

3. $\implies$ 1. : Since λ^* is a Carathéorody measure, then all Borel sets is measurable. Specially, the G_δ set U is measurable. And since λ^* -zero set is measurable, then

$$E = U \setminus (U \setminus E)$$

is measurable.

2. $\implies$ 4. : There is an open set $U \supseteq E^c$ such that $\lambda^*(U \setminus E^c) < \varepsilon$. Let $F = U^c$, then

$$E \setminus F = E \cap F^c = E \cap U = U \setminus E^c$$

Thus

$$\lambda^*(E \setminus F) = \lambda^*(U \setminus E^c) < \varepsilon$$

4. $\implies$ 5. : For each $n \in \mathbb{Z}_{>0}$, there is a closed set $F_n \subseteq E$ such that

$$\lambda^*(E \setminus F_n) < \frac{1}{n}$$

Let

$$F := \bigcup_{n=1}^{\infty} F_n,$$

then F is a F_σ set, such that

$$\lambda^*(E \setminus F) \leq \lambda^*(E \setminus F_n) < \frac{1}{n}, \quad \forall n.$$

That is $\lambda^*(E \setminus F) = 0$.

5. $\implies$ 1. : Since $E \setminus F$ and F is measurable, we have

$$E = (E \setminus F) \cup F$$

is measurable. □

Exercises for Lebesgue Measure

Exercise 1.3.10

With λ_t defined by

$$\lambda_t(E) = \lambda(|t|E),$$

prove that $\lambda_t(N) = 0$ whenever $\lambda(N) = 0$.

Proof. Since $\lambda(N) = 0$, for each ε , there is a sequence of closed intervals $I_1, \dots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \frac{\varepsilon}{|t|^n}$$

where n is the dimension of the Euclidean space we discuss. Then

$$|t| E \subseteq \bigcup_{k=1}^{\infty} |t| I_k, \quad \sum_{k=1}^{\infty} v(|t| I_k) = |t|^n \sum_{k=1}^{\infty} v(I_k) < \varepsilon$$

That is

$$\lambda^*(|t| E) < \varepsilon, \quad \forall \varepsilon > 0$$

Thus $|t| E$ is measurable, and

$$\lambda(|t| E) = \lambda^*(|t| E) = 0$$

□

Exercise 1.3.11

Let $I, I_1, I_2, \dots, I_k$ be intervals in $\mathbb{R}$ such that $I \subset \bigcup_{i=1}^k I_i$. Prove that

$$v(I) \leq \sum_{i=1}^k v(I_i),$$

where $v(I)$ denotes the length of the interval I .

Proof. We may assume that $v(I) < \infty$, that is I is bounded. Since $v(I) = v(\bar{I})$, $v(I_i) = v(\bar{I}_i)$, and $\bar{I} \subseteq \bigcup_{i=1}^k \bar{I}_i$, we may assume that $I, I_1, \dots, I_k$ are closed intervals. For any $\varepsilon > 0$, there are closed intervals I'_i such that

$$I_i \subseteq (I'_i)^\circ, \quad v(I'_i) \leq v(I_i) + \frac{\varepsilon}{2^i}$$

Then $\{(I'_i)^\circ\}$ is a open cover of I . Since I is compact, it has a Lebesgue number η . We can subdivide I into subintervals $K_1, \dots, K_m$ with diameter less than η . Then each K_j is contained in some $(I'_{i_j})^\circ$.

We can formalize this by defining a map $f : \{1, \dots, m\} \rightarrow \{1, \dots, k\}$ such that for each j , $K_j \subset (I'_{f(j)})^\circ$. Since the subintervals K_j have disjoint interiors and their union is I , we can write the total length of I and group the sum

according to the covering interval index:

$$v(I) = \sum_{j=1}^m v(K_j) = \sum_{i=1}^k \left(\sum_{j: f(j)=i} v(K_j) \right).$$

For any fixed i , the intervals $\{K_j\}_{j: f(j)=i}$ are a collection of disjoint intervals all contained within $(I'_i)^\circ$. Therefore, the sum of their lengths cannot exceed the length of $(I'_i)^\circ$:

$$\sum_{j: f(j)=i} v(K_j) = v \left(\bigcup_{j: f(j)=i} K_j \right) \leq v((I'_i)^\circ) = v(I'_i).$$

Applying this inequality to our grouped sum, we obtain a single chain of inequalities:

$$v(I) = \sum_{i=1}^k \left(\sum_{j: f(j)=i} v(K_j) \right) \leq \sum_{i=1}^k v(I'_i) \leq \sum_{i=1}^k \left(v(I_i) + \frac{\epsilon}{2^i} \right) = \sum_{i=1}^k v(I_i) + \epsilon \sum_{i=1}^k \frac{1}{2^i}.$$

Since $\sum_{i=1}^k \frac{1}{2^i} < 1$, we have $v(I) < \sum_{i=1}^k v(I_i) + \epsilon$. As $\epsilon > 0$ was arbitrary, we conclude that

$$v(I) \leq \sum_{i=1}^k v(I_i).$$

□

Exercise 1.3.12

Let $E \subset \mathbb{R}$, and for each real number t , let $E + t = \{x + t : x \in E\}$. Prove that $\lambda^*(E) = \lambda^*(E + t)$. From this show that if E is Lebesgue measurable, then so is $E + t$.

Proof. Once we show that

$$\lambda^*(E + t) \leq \lambda^*(E), \quad \forall t$$

. By replacing t by $-t$, E by $E + t$, we have

$$\lambda^*(E) \leq \lambda^*(E + t), \quad \forall t$$

Then we get $\lambda^*(E) = \lambda^*(E + t)$. To show this, for each $\varepsilon > 0$, we take closed intervals $I_1, \dots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \lambda(E) + \varepsilon$$

Then

$$E + t \subseteq \bigcup_{k=1}^{\infty} (I_k + t), \quad \sum_{k=1}^{\infty} v(I_k + t) = \sum_{k=1}^{\infty} v(I_k) < \lambda(E) + \varepsilon.$$

Thus

$$\lambda(E + t) \leq \lambda(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, $\lambda(E + t) \leq \lambda(E)$.

To show that $E + t$ is Lebesgue measurable. There is a G_δ set G , such that

$$G \supseteq E, \quad \lambda^*(G \setminus E) = 0.$$

Since open sets is closed under translation and since intersection is kept, then $G_\delta + t$ is a G_δ set as well. We have

$$(G + t) \supseteq (E + t), \quad (G + t) \setminus (E + t) = (G \setminus E) + t$$

Thus

$$\lambda^*((G + t) \setminus (E + t)) = \lambda^*((G \setminus E) + t) = \lambda^*(G \setminus E) = 0$$

Then it follows from Theorem 1.3.9 that $E + t$ is measurable. □

Exercise 1.3.13 ▶ *

Prove that Lebesgue measure on $\mathbb{R}^n$ is independent of the choice of coordinate system. That is, prove that Lebesgue outer measure is invariant under rigid motions in $\mathbb{R}^n$.

Remark.

或许需要积分的替换公式, 之后再来看.

Proof. Since we have shown that Lebesgue outer measure is invariant under translation, we only need to show that it is also invariant under Orthogonal Transformation. □

Exercise 1.3.14

Let P denote an arbitrary $(n - 1)$ -dimensional hyperplane in $\mathbb{R}^n$. Prove that $\lambda(P) = 0$.

Proof. Since Lebesgue measure is invariant under rigid motions. We may assume that $P = \{x_n = 0\}$. For each $\varepsilon > 0$, let

$$I_k := \left[-\frac{k}{2}, \frac{k}{2}\right]^{n-1} \times \left[\frac{-\varepsilon}{2k^{n-1}} \cdot \frac{1}{2^k}, \frac{\varepsilon}{2k^{n-1}} \cdot \frac{1}{2^k}\right]$$

Then

$$P \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \nu(I_k) = \frac{\varepsilon}{2^k}, \quad \forall k$$

Thus

$$\lambda(P) \leq \sum_{k=1}^{\infty} \nu(I_k) \leq \varepsilon$$

Since ε is arbitrary, $\lambda(P) = 0$. □

Exercise 1.3.15

In this problem, we want to show that every Lebesgue measurable subset of $\mathbb{R}$ must be “densely populated” in some interval. Thus, let $E \subset \mathbb{R}$ be a Lebesgue measurable set, $\lambda(E) > 0$. For each $\varepsilon > 0$, show that there exists an interval I such that

$$\frac{\lambda(E \cap I)}{\lambda(I)} > 1 - \varepsilon.$$

Note.

我们发现构造这样一个 I 是及其困难的. 转而使用反证法, 去论证 “如果一个建筑的每一块砖都不合格, 那么整个建筑的质量必然有问题”.

Proof. Suppose conversely, there exists an $\varepsilon > 0$, such that for each interval I , we have

$$\frac{\lambda(E \cap I)}{\lambda(I)} \leq 1 - \varepsilon.$$

Let U be a open set such that

$$U \supseteq E, \quad \lambda(U \setminus E) < \delta.$$

Suppose

$$U = \bigcup_{k=1}^{\infty} I_k$$

where I_k are disjoint open intervals. Then

$$\lambda(E \cap U) = \sum_{k=1}^{\infty} \lambda(E \cap I_k) \leq (1 - \varepsilon) \sum_{k=1}^{\infty} \lambda(I_k) = (1 - \varepsilon) \lambda(U)$$

Since

$$\lambda(U) = \lambda(E \cap U) + \lambda(U \setminus E)$$

, we have

$$\lambda(E \cap U) + \lambda(U \setminus E) \geq \frac{1}{1 - \varepsilon} \lambda(E \cap U)$$

Thus

$$\delta \geq \left(1 - \frac{1}{1 - \varepsilon}\right) \lambda(E \cap U) = \frac{\varepsilon}{1 - \varepsilon} \lambda(E)$$

since $E \cap U = E$. Note that δ can be chosen arbitrarily and independent with ε and $\lambda(E)$, then it leads to a contradiction, which completes the proof. $\square$

Exercise 1.3.16

Prove that every set $A \subset \mathbb{R}^n$ is contained within a G_δ -set G with the property $\lambda(G) = \lambda^*(A)$.

Proof. For any $n > 0$, there are closed intervals $I_1^{(n)}, I_2^{(n)}, \dots$ such that

$$A \subseteq \bigcup_{k=1}^{\infty} I_k^{(n)}, \quad \sum_{k=1}^{\infty} v(I_k^{(n)}) \leq \lambda^*(A) + \frac{1}{2n}$$

For each k , there is a closed interval $(I_k^{(n)})'$ such that

$$I_k^{(n)} \subseteq \left((I_k^{(n)})'\right)^\circ, \quad v\left((I_k^{(n)})'\right) < v(I_k^{(n)}) + \frac{1}{2^{k+1}n}$$

Let

$$G := \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} \left((I_k^{(n)})' \right)^{\circ}$$

is a G_{δ} set. Then

$$\lambda(G) \leq \lambda \left(\bigcup_{k=1}^{\infty} (I_k^{(n)})' \right) \leq \sum_{k=1}^{\infty} v \left((I_k^{(n)})' \right) < \lambda^*(A) + \frac{1}{n}, \quad \forall n$$

Thus

$$\lambda(G) \leq \lambda^*(A)$$

The other hand, since $G \supseteq A$, it is immediate that

$$\lambda(G) = \lambda^*(G) \geq \lambda^*(A).$$

Finally,

$$\lambda(G) = \lambda^*(A)$$

□

Exercise 1.3.17

Let $\{E_k\}$ be a sequence of Lebesgue measurable sets contained in a compact set $K \subset \mathbb{R}^n$. Assume for some $\varepsilon > 0$ that $\lambda(E_k) > \varepsilon$ for all k . Prove that there is some point that belongs to infinitely many E_k 's.

Note.

对于由一系列成员组成的族类, 一个点属于无穷多个成员, 当且仅当它永远无法仅被有限多个成员“杀死”, 当且仅当它是“反复出现”的, 当且仅当它属于集合列的极限上集.

Proof. A point belongs to infinitely many E_k , if and only if it belongs to

$$\limsup_{k \rightarrow \infty} E_k := \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$$

Since $\lambda \left(\bigcup_{k=1}^{\infty} E_k \right) \leq \lambda(K) < \infty$. It follows from Corollary 1.1.11 that

$$\lambda \left(\limsup_{k \rightarrow \infty} E_k \right) \geq \limsup_{k \rightarrow \infty} \lambda(E_k) \geq \varepsilon > 0$$

Thus

$$\limsup_{k \rightarrow \infty} E_k \neq \emptyset$$

which completes the proof. $\square$

Exercise 1.3.18

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a Lipschitz map. Prove that if $\lambda(E) = 0$, then $\lambda(T(E)) = 0$.

Proof. Suppose that

$$d(T(x_1), T(x_2)) \leq L \cdot d(x_1, x_2)$$

Since $\lambda(E) = 0$, for any $\varepsilon > 0$, there are closed intervals $I_1, \dots$, such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \varepsilon$$

There are finitely many Cubes $Q_{k1}, \dots, Q_{km_k}$, such that

$$I_k \subseteq \bigcup_{i=1}^{m_k} Q_{ki}, \quad \sum_{i=1}^{m_k} v(Q_{ki}) < 2v(I_k)$$

Thus we may assume that I_k are all cubes. Denote the diameter of I_k by $\sqrt{n}a_k$, and denote the centre of I_k by c_k , then

$$d(T(x), T(c_k)) \leq L \cdot d(x, c_k) \leq \sqrt{n}L \cdot a_k$$

Thus $T(I_k)$ is contained by the cube centering $T(c_k)$ with diameter $nL \cdot a_k$, which we denote by I'_k , then

$$v(I'_k) \leq n^n L^n v(I_k)$$

. We have

$$T(E) \subseteq \bigcup_{k=1}^{\infty} I'_k, \quad \sum_{k=1}^{\infty} v(I'_k) < n^n L^n \varepsilon$$

, which concludes that $\lambda(T(E)) = 0$. $\square$

1.4 The Cantor Set

Definition 1.4.1 ► Cantor set

The Cantor set is a subset of the interval $[0, 1]$. We will describe it by constructing its complement in $[0, 1]$.

1. At the first step, let $I_{1,1}$ denote the open interval $\left(\frac{1}{3}, \frac{2}{3}\right)$. Thus $I_{1,1}$ is the open middle third of the interval $I = [0, 1]$.
2. The second step involves performing the first step on each of the two remaining intervals of $I - I_{1,1}$. That is, We produce two open intervals, $I_{2,1}$ and $I_{2,2}$, each being the open middle third of one of the two intervals that constitute $I - I_{1,1}$.
3. At the i th step we produce 2^{i-1} open intervals $I_{i,1}, I_{i,2}, \dots, I_{i,2^{i-1}}$, each of length $\left(\frac{1}{3}\right)^i$. The $(i + 1)$ th step consists in producing middle thirds of each of the intervals of

$$I - \bigcup_{j=1}^i \bigcup_{k=1}^{2^{j-1}} I_{j,k}.$$

4. With C denoting the Cantor set, we define its complement by

$$I \setminus C = \bigcup_{j=1}^{\infty} \bigcup_{k=1}^{2^{j-1}} I_{j,k}.$$

Proposition 1.4.2

Let C be the Cantor set. Then the following holds:

1. C is closed, that is $C = \bar{C}$.
2. Let λ be the Lebesgue measure, then

$$\lambda(\bar{C}) = \lambda(C) = 0$$

3. C is nowhere dense.
4. The cardinality of C is

$$|C| = 2^{\aleph_0} = |R| = c$$

Proof. 1. $I \setminus C$ is open by definition. Thus C is closed.

2.

$$\lambda(I \setminus C) = \sum_{j=1}^{\infty} \sum_{k=1}^{2^{j-1}} \lambda(I_{j,k}) = \sum_{j=1}^{\infty} 2^{j-1} \left(\frac{1}{3}\right)^j = \frac{1}{2} \sum_{j=1}^{\infty} \left(\frac{2}{3}\right)^j = 1$$

Thus

$$\lambda(C) = 1 - \lambda(I \setminus C) = 1 - 1 = 0$$

Note.

$[0, 1]$ 上的每一个值通过修饰存在唯一的二进制表示. 同样地也存在唯一的三进制表示. Cantor 集就是把每个三进制表示中, 字符含 1 的数都去掉.

3.

Every number $x \in [0, 1]$ has a ternary expansion of the form

$$x = \sum_{i=1}^{\infty} \frac{x_i}{3^i}$$

where each x_i is 0, 1, or 2 and we write $x = 0.x_1x_2\cdots$. This expansion is unique except when

$$x = \frac{a}{3^n}$$

(Since $\frac{1}{3^n}$ has another expression $\sum_{i=n+1}^{\infty} \frac{2}{3^i}$) where a and n are positive integers with $0 < a < 3^n$ and where 3 does not divide a . In this case, x has the form

$$x = \sum_{i=1}^n \frac{x_i}{3^i}$$

where x_i is either 1 or 2. If $x_n = 2$, we will use this expression to represent x . However, if $x_n = 1$, we will use the following representation for x :

$$x = \frac{x_1}{3} + \frac{x_2}{3^2} + \cdots + \frac{x_{n-1}}{3^{n-1}} + \frac{0}{3^n} + \sum_{i=n+1}^{\infty} \frac{2}{3^i}.$$

Thus, with this convention, each number $x \in [0, 1]$ has a unique ternary expansion.

Let $x \in I$ and consider its ternary expansion $x = 0.x_1x_2\cdots$, bearing in mind the convention we have adopted above. Observe that $x \notin I_{1,1}$ if and only if $x_1 \neq 1$. Also, if $x_1 \neq 1$, then $x \notin I_{2,1} \cup I_{2,2}$ if and only if $x_2 \neq 1$. Continuing in this way, we see that $x \in C$ if and only if $x_i \neq 1$ for each positive integer i . Thus, there is a one-to-one correspondence between

elements of C and all sequences $\{x_i\}$ in which each x_i is either 0 or 2. The cardinality of the latter is $2^{\aleph_0}$, which is $|\mathbb{R}| = c$.

□

Exercises for Cantor set

Exercise 1.4.3

Prove that the Cantor-type set C_α (The method of construction is the same as C , except that at the i th step, each of the intervals removed has length $\alpha 3^{-i}$) is nowhere dense, has cardinality c , and has Lebesgue measure $1 - \alpha$.

Proof. • $\overline{C}_\alpha = C_\alpha$. Only need to show that

$$\text{int}(C_\alpha) = \emptyset$$

With E_k denoting the result of the k th step constructing the C_α , we define its complements by

$$I \setminus E_k := \bigcup_{j=1}^k \bigcup_{m=1}^{2^{j-1}} I_{m,j}$$

Then $E_1 \supseteq E_2 \supseteq \dots$, and $C_\alpha = \bigcap_{k=1}^{\infty} E_k$. If there is a nonempty open interval $(a, b) \subseteq C_\alpha$. Then

$$(a, b) \subseteq E_k, \quad \forall k$$

Note that E_k is a disjoint union of closed intervals with length l_k , which satisfies

$$l_k = \frac{1}{2} (l_{k-1} - \alpha 3^{-k}) < \frac{1}{2} l_{k-1} \implies l_k < \frac{1}{2^k}$$

. There is a k_0 such that $l_{k_0} < b - a$. Since

$$(a, b) \subseteq E_{k_0}$$

is a connected set lying in E_{k_0} , E_{k_0} has a component with length more than $b - a$, which is a contradiction. Thus there is no open interval contained in C_α , that is $C_\alpha = \overline{C}_\alpha$ is nowhere dense.

- We denote the two closed intervals of $I \setminus I_{1,1}$ by J_1, J_2 . Denote the two closed intervals of $J_1 \setminus I_{2,1}$ by $J_{1,1}, J_{1,2}$, and denote the two that of $J_2 \setminus I_{2,2}$ by $J_{2,1}, J_{2,2}$. Inductively, denote the two intervals of $J_{i_1, i_2, \dots, i_{j-1}} \setminus I_{j, 2^{j-1}i_1 + \dots + 2^0 i_{j-1}}$ by $J_{i_1, \dots, i_{j-1}, 1}, J_{i_1, \dots, i_{j-1}, 2}$. Then

$$E_k = \bigcup_{i_1, \dots, i_k \in \{1, 2\}^k} J_{i_1, \dots, i_k}$$

Let

$$\Sigma := \{1, 2\}^{\mathbb{N}}$$

Given a $(s_1, s_2, \dots) \in \Sigma$, we can get a sequence of nested closed intervals

$$J_{s_1} \supseteq J_{s_1, s_2} \supseteq \dots$$

From Nested Interval Theorem, there is a unique point $s \in \bigcap_{k=1}^{\infty} J_{s_1, \dots, s_k}$. We define a f by mapping $(s_1, s_2, \dots) \in \Sigma$ to $s \in C_\alpha$. To show that f is injective, we take

$$s = (s_1, s_2, \dots), t = (t_1, t_2, \dots) \in \Sigma, \quad s \neq t$$

Then there is a smallest $k \in \mathbb{Z}_{>0}$, such that $s_k \neq t_k$. We may assume that $s_k = 1, t_k = 2$. Then

$$f(s) \in J_{s_1, s_2, \dots, s_{k-1}, 1}, \quad f(t) \in J_{t_1, t_2, \dots, t_{k-1}, 2} = J_{s_1, s_2, \dots, s_{k-1}, 2}$$

Since $J_{s_1, s_2, \dots, s_{k-1}, 1} \cap J_{s_1, s_2, \dots, s_{k-1}, 2} = \emptyset$, $f(s) \neq f(t)$. Thus f is injective. To show that f is surjective. Take any $x \in C_\alpha$. Since

$$C_\alpha = \bigcap_{k=1}^{\infty} E_k$$

, we have

$$x \in E_k, \quad \forall k$$

And since

$$E_1 = J_1 \cup J_2$$

is a disjoint union. There exists a unique $i_1 \in \{1, 2\}$, such that

$$x \in J_{i_1}$$

Then

$$x \in E_2 \cap J_{i_1} = J_{i_1,1} \cup J_{i_1,2}$$

There exists a unique $i_2 \in \{1, 2\}$ such that

$$x \in J_{i_1, i_2}$$

Inductively, once we show that there exists a unique $(i_1, \dots, i_k) \in \{1, 2\}^k$ such that

$$x \in J_{i_1, i_2, \dots, i_k}$$

From

$$x \in E_{k+1} \cap J_{i_1, \dots, i_k} = J_{i_1, \dots, i_k, 1} \cup J_{i_1, \dots, i_k, 2}$$

we know that there exists a unique $(i_1, \dots, i_{k+1}) \in \{1, 2\}^{k+1}$, such that

$$x \in J_{i_1, \dots, i_{k+1}}$$

Thus we have determined a unique $(i_1, i_2, \dots) \in \Sigma$, such that

$$x \in \bigcap_{k=1}^{\infty} J_{i_1, i_2, \dots, i_k}$$

That is

$$f(i_1, i_2, \dots) = x$$

Thus f is surjective. Finally, we have

$$c = 2^{\aleph_0} = |\Sigma| = |C_\alpha|$$

•

$$\lambda(I \setminus C_\alpha) = \lambda\left(\bigcup_{j=1}^{\infty} \bigcup_{k=1}^{2^{j-1}} I_{k,j}\right) = \sum_{j=1}^{\infty} 2^{j-1} \alpha \left(\frac{1}{3}\right)^j = \alpha$$

Thus

$$\lambda(C_\alpha) = \lambda(I) - \lambda(I \setminus C_\alpha) = 1 - \alpha$$

□

Exercise 1.4.4

Construct an open set $U \subset [0, 1]$ such that U is dense in $[0, 1]$, $\lambda(U) < 1$, and $\lambda(U \cap (a, b)) > 0$ for every interval $(a, b) \subset [0, 1]$.

Proof. Define

$$U := I \setminus C_\alpha, \quad 0 < \alpha < 1$$

Where C_α is the Cantor type set with Lebesgue measure $1 - \alpha$. Then

$$\lambda(U) = \lambda(I) - \lambda(C_\alpha) = \alpha > 0$$

If U is not dense. Then there is a nonempty open subset $Y \subseteq [0, 1]$ such that

$$Y \subseteq U^c = C_\alpha$$

which is a contradiction to C_α is nowhere dense. Since U is dense,

$$U \cap (a, b) \neq \emptyset$$

That is $U \cap (a, b)$ is a nonempty open set, thus its Lebesgue measure is nonzero. $\square$

Exercise 1.4.5

Prove that the family of Borel subsets of $\mathbb{R}$ has cardinality c . From this deduce the existence of a Lebesgue measurable set that is not a Borel set.

Exercise 1.4.6

Let E be the set of numbers in $[0, 1]$ whose ternary expansions have only finitely many 1's. Prove that $\lambda(E) = 0$.

Proof. Let E_k be the set of numbers in $I = [0, 1)$ whose ternary expansions have no 1's after the k th digit. Then

$$E = \bigcup_{k=1}^{\infty} E_k, \quad E_1 \subseteq E_2 \subseteq \dots$$

Let I_1, I_2, I_3 be the former, middle, latter third of the $I = [0, 1)$ respectively, specially, we say $[0, \frac{1}{3})$, $[\frac{1}{3}, \frac{2}{3})$, $[\frac{2}{3}, 1)$. Once we have defined $I_{i_1, i_2, \dots, i_k}$, then we define $I_{i_1, i_2, \dots, i_k, 1}$ by the first third of the interval $I_{i_1, i_2, \dots, i_k}$ as before, the

remainings are similar. We have

$$E_k = I \setminus \bigcup_{j=k}^{\infty} \bigcup_{i_1, i_2, \dots, i_k \in \{1, 2, 3\}^k, i_{k+1}, \dots, i_j \in \{1, 3\}^{j-k}} I_{i_1, i_2, \dots, i_j, 2}$$

Let

$$A_{k,m} := \bigcup_{j=k}^m \bigcup_{i_1, i_2, \dots, i_k \in \{1, 2, 3\}^k, i_{k+1}, \dots, i_j \in \{1, 3\}^{j-k}} I_{i_1, i_2, \dots, i_j, 2}$$

Then

$$A_{k,k} \subseteq A_{k,k+1} \subseteq \dots$$

Denote $\bigcup_{m=k}^{\infty} A_{k,m}$ by A_k , then $E_k = I \setminus A_k$. We have

$$\lambda(A_k) = \lim_{m \rightarrow \infty} \lambda(A_{k,m}) = \sum_{j=k}^{\infty} 3^k 2^{j-k} \left(\frac{1}{3}\right)^{j+1} = \frac{1}{3} \left(\frac{3}{2}\right)^k \sum_{j=k}^{\infty} \left(\frac{2}{3}\right)^j = 1$$

Thus

$$\lambda(E_k) = \lambda(I) - \lambda(A_k) = 1 - 1 = 0$$

It follows from the continuity of λ from below that

$$\lambda(E) = \lim_{k \rightarrow \infty} \lambda(E_k) = 0$$

□

Proof. Another proof, use Cantor set.

Note.

- 标准 Cantor 集合描述了 $[0,1]$ 内三进制表示不含 1 的那些数。
- 通过对 Cantor 集 $\frac{1}{3^k}$ 倍的缩放, 可以实现所描述的三进制范围表示相对于小数点的滞后 (k 个字符)。
- 缩放后 Cantor 集的起点, 决定了新 Cantor 集开始作用的之前的三进制字符都是哪些。
- 一切的一切, 源于 Cantor 集的自相似性, 也就是说

$$C = \left(\frac{1}{3}C\right) + \left(\frac{1}{3}C + \frac{2}{3}\right)$$

E_k is defined as above. Let $P_k := \{0, 1, 2\}^k$. For $p = (d_1, \dots, d_k) \in P_k$, define

$$C_p := \frac{1}{3^k} C + \sum_{j=1}^k \frac{d_j}{3^j}$$

Then the similar process we describe C gives that

$$C_p = \left\{ x \in [0, 1] : x = (0.m_1 m_2 \dots)_{\text{ternary}}, m_i = d_i \text{ for } 1 \leq i \leq k, m_j \neq 1 \text{ for } j > k \right\}$$

We have

$$E_k = \bigcup_{p \in P_k} C_p$$

Since

$$\lambda(C_p) = \frac{1}{3^k} \lambda(C) = 0$$

We have

$$E_k \leq \sum_{p \in P_k} \lambda(C_p) = 0$$

Then

$$\lambda(E) = \lim_{k \rightarrow \infty} \lambda(E_k) = 0$$

□